

Experiment-5 Write the python program for missionaries cannibal problem

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vaccum cleaner.py - C:/Users/prana/OneDrive/Desktop/vaccum cleaner.py (3.13.14)
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from collections import deque
|
start = (3, 3, 0)
goal = (0, 0, 1)

moves = [(1,0), (2,0), (0,1), (0,2), (1,1)]

def safe(m, c):
    if m < 0 or c < 0 or m > 3 or c > 3:
        return False

    if m > 0 and m < c:
        return False

    if (3-m) > 0 and (3-m) < (3-c):
        return False

    return True

queue = deque([(start, [])])
visited = {start}

while queue:
    state, path = queue.popleft()

    if state == goal:
        print("Solution:")
        for s in path:
            print(s)
        break

    m, c, boat = state

    for dm, dc in moves:
        if boat == 0:
            new = (m-dm, c-dc, 1)
        else:
            new = (m+dm, c+dc, 0)

        if safe(new[0], new[1]) and new not in visited:
            visited.add(new)
            queue.append((new, path + [new]))
```

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IDLE Shell 3.13.14
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Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/prana/OneDrive/Desktop/vaccum cleaner.py =====
Solution:
(3, 1, 1)
(3, 2, 0)
(3, 0, 1)
(3, 1, 0)
(1, 1, 1)
(2, 2, 0)
(0, 2, 1)
(0, 3, 0)
(0, 1, 1)
(1, 1, 0)
(0, 0, 1)
>>> |
```